

Hadamard gate: $\tilde{H}$, $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$

$$\tilde{H} \begin{pmatrix} |0\rangle \\ |1\rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} |0\rangle \\ |1\rangle \end{pmatrix}$$

$$\Rightarrow \tilde{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

• $\tilde{H}^\dagger = H$ and $H^\dagger H = \mathbb{1} = H^2$

$$H V = \lambda V$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = 0 \quad \begin{matrix} \lambda_1 = \frac{1}{\sqrt{2}} \\ \lambda_2 = -\frac{1}{\sqrt{2}} \end{matrix}$$

$$V_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$V_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\tilde{H} V_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$H V_2 = - \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$H = |0\rangle\langle 0| + |1\rangle\langle 0| + |0\rangle\langle 1| - |1\rangle\langle 1|$$

Hamiltonian has a spectral decomp: $\tilde{H} = \sum_E E |E\rangle\langle E|$

$$\Sigma \equiv \sigma_z = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

• Suppose $H = \hbar\omega \Sigma$

then $(|0\rangle + |1\rangle)/\sqrt{2}$ and $(|0\rangle - |1\rangle)/\sqrt{2}$ are eigenstates with energies $+\hbar\omega$, $-\hbar\omega$

$\Rightarrow$ Ground state: $(|0\rangle - |1\rangle)/\sqrt{2}$

$$|\psi(t_2)\rangle = \exp(-i H (t_2 - t_1)/\hbar) |\psi(t_1)\rangle$$

$$U(t_1, t_2) \equiv \exp(\hat{A})$$

Entanglement: 2-qubit state $|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$

* There is no single qubit states $|a\rangle$ and $|b\rangle$ s.t.

$$|\psi\rangle = |a\rangle|b\rangle$$

Proof: Suppose $|a\rangle = \alpha|0\rangle + \beta|1\rangle$

$$|b\rangle = \gamma|0\rangle + \delta|1\rangle$$

$$|a\rangle|b\rangle = \alpha\gamma|0\rangle|0\rangle + \alpha\delta|0\rangle|1\rangle + \beta\gamma|1\rangle|0\rangle + \beta\delta|1\rangle|1\rangle$$

$$\therefore \alpha\delta = 0, \beta\gamma = 0$$

$$\alpha = 0 \text{ then } |0\rangle|0\rangle = 0$$

$\delta = 0$ then $|1\rangle|1\rangle = 0$
 so no such $|a\rangle|b\rangle$ exist.

Super dense coding:

- Goal: transmit info from A to B Classically

A: Has two classical bits of info

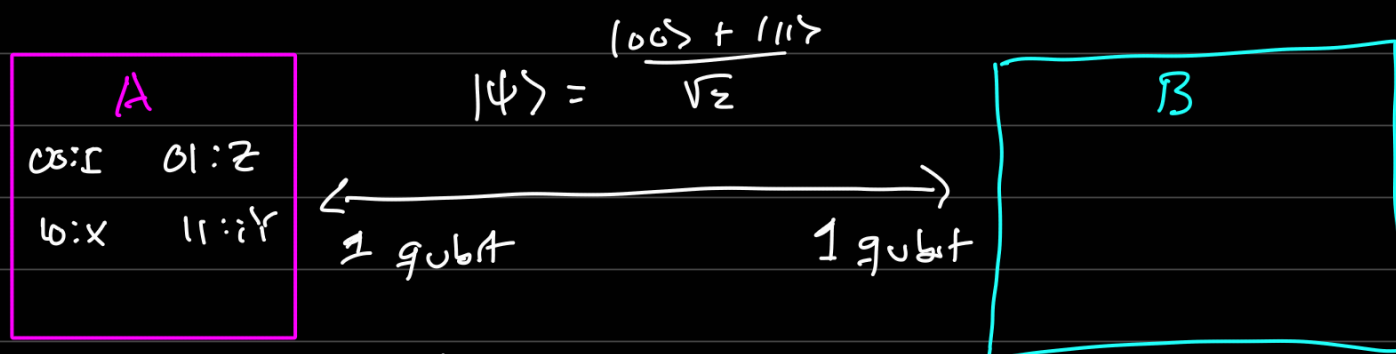
→ Can only send a single qubit.

- Suppose A & B initially share a pair of qubits in an entangled state $|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$

$$= \frac{|0\rangle|0\rangle + |1\rangle|1\rangle}{\sqrt{2}}$$

A: possesses the first qubit

B: possesses the second



- By sending A's qubit to B, A can communicate two bits of classical info

If A wants to send some info her actions

act on $|\psi\rangle$ as $(U_A \otimes I_B)$

(i) $|00\rangle$: I

(ii) $|01\rangle$: $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$(Z_A \otimes I_B)|\psi\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

phase flip

(iii) $|10\rangle$: $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$= \frac{1}{\sqrt{2}} (|11\rangle + |01\rangle)$$

(iv) $|11\rangle$: $iY = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

$$= \frac{1}{\sqrt{2}} (-|10\rangle + |01\rangle)$$

* Classical bits: $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 I Z X iY

This mapping is chosen

EPR pairs (Bell basis, Bell states):

$$00: |\psi\rangle \rightarrow (|00\rangle + |11\rangle) / \sqrt{2}$$

$$01: |\psi\rangle \rightarrow (|00\rangle - |11\rangle) / \sqrt{2}$$

$$10: |\psi\rangle \rightarrow (|10\rangle + |01\rangle) / \sqrt{2}$$

$$11: |\psi\rangle \rightarrow (|10\rangle - |01\rangle) / \sqrt{2}$$

1
2
3
4

$$|\psi\rangle = \sum \alpha_i |\tilde{\psi}_i\rangle$$

- They span a basis of $\mathcal{H}_1 \otimes \mathcal{H}_2$

Density operator:

(Ensembles of quantum systems)

- Good for describing quantum systems whose state is not completely known.

- Suppose a quantum system is in one of a number of states $|\psi_i\rangle$ with probability p_i :

$\{p_i, |\psi_i\rangle\}$: ensemble of pure states

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i| \quad (\text{density operator})$$

- $\psi_i \rightarrow U |\psi_i\rangle$ then

$$\rho \xrightarrow{U} \sum_i p_i U^\dagger |\psi_i\rangle \langle \psi_i| U = U^\dagger \rho U$$

- For a measurement: $\{M_m\}$

$$p(m|i) = \langle \psi_i | M_m^\dagger M_m | \psi_i \rangle = \text{Tr}(M_m^\dagger M_m |\psi_i\rangle \langle \psi_i|)$$

Since

$$\begin{aligned} \text{Tr}(M_m^\dagger M_m |\psi_i\rangle \langle \psi_i|) &= \sum_k \langle k | M_m^\dagger M_m | \psi_i \rangle \langle \psi_i | k \rangle \\ &= \sum_k \langle \psi_i | k \rangle \langle k | M_m^\dagger M_m | \psi_i \rangle \\ &= \langle \psi_i | \sum_k | k \rangle \langle k | M_m^\dagger M_m | \psi_i \rangle \\ &= \langle \psi_i | M_m^\dagger M_m | \psi_i \rangle \end{aligned}$$

- For independent variables m, i the total prob.

$p(m, i)$: partials by

$$p(m) = \sum_i p(m, i) \quad p(i) = \sum_m p(m, i)$$

$$p(i) = \sum_m P(m, i) \quad \sum_i P(m, i) = \sum_i P(m, i)$$

• Conditional prob. $P(m|i) = P(m, i) / p(i)$

$$\begin{aligned} p(m) &= \sum_i P(m, i) = \sum_i P(m|i) p(i) \\ &= \sum_i \langle \psi_i | M_m^\dagger M_m | \psi_i \rangle p(i) \\ &= \sum_i \text{Tr}(M_m^\dagger M_m (p(i) |\psi_i\rangle \langle \psi_i|)) \end{aligned}$$

$$p(m) = \text{Tr}(M_m^\dagger M_m \rho)$$

* For a measured state $|\psi_i^m\rangle = \frac{M_m |\psi_i\rangle}{\sqrt{\langle \psi_i | M_m^\dagger M_m | \psi_i \rangle}}$

What is the probability of having $|\psi_i^m\rangle$?

$$P(m|i) = \langle \psi_i | M_m^\dagger M_m | \psi_i \rangle$$

$$\sum_i \frac{\tilde{P}_{m,i}}{P(m|i)} M_m^\dagger |\psi_i\rangle \langle \psi_i| M_m$$

$$\tilde{P}_{m,i} = P(m|i) = \frac{P(m, i)}{p(m)} = \frac{P(m|i) p(i)}{p(m)}$$

$$\rho_m = \sum_i \frac{p_i}{\text{Tr}(M_m^\dagger M_m \rho)} M_m^\dagger |\psi_i\rangle \langle \psi_i| M_m = \frac{M_m^\dagger \rho M_m}{\text{Tr}(M_m^\dagger M_m \rho)}$$

$$\rho_m = \frac{M_m^\dagger \rho M_m}{\text{Tr}(M_m^\dagger M_m \rho)}$$

Def: A pure state: a Quantum system whose state $|\psi\rangle$ is known exactly.

$$\rho = |\psi\rangle \langle \psi|$$

Mixed state: otherwise.

* A pure state satisfies: $\text{Tr}(\rho^2) = 1$

A Mixed " " " " $\text{Tr}(\rho^2) < 1$

• Suppose we prepare a state ρ_i with prob p_i

$$\rho_i = |\psi_i\rangle \langle \psi_i| \quad (\text{pure})$$

$$\Rightarrow \text{System: } \sum_i p_i \rho_i$$

Proof: Suppose ρ_i arises from $\{p_{ij}, |\psi_{ij}\rangle\}$

$$\rho_i = \sum_j p_{ij} |\psi_{ij}\rangle \langle \psi_{ij}|$$

• probability of being in $|\psi_{ij}\rangle$

• probability of ρ_i is p_i so the prob

$$P(|\psi_{ij}\rangle) = p_i p_{ij}$$

$$\begin{aligned} \rho &= \sum_i (p_i p_{ij}) |\psi_{ij}\rangle \langle \psi_{ij}| = \sum_i p_i \sum_j p_{ij} |\psi_{ij}\rangle \langle \psi_{ij}| \\ &= \sum_i p_i \rho_i \end{aligned}$$

* ρ is a mixture of states ρ_i with $P_i = P(\rho_i)$

• Suppose we have a system which we measure S .
Each measurement we apply m measurements. The state will end up at ρ_m with prob $P(m)$

• Assuming it was prepared at ρ_i we would have no idea if we "lost" the data. Don't know the value of m

$$\rho = \sum_m P(m) \rho_m, \quad \left| \begin{aligned} P(m) &= \sum_i P(m|i) P(i) \\ &= \text{tr}(M_m^\dagger M_m \rho) \end{aligned} \right.$$

$$\rho_m = M_m \rho M_m^\dagger / \text{tr}(M_m^\dagger M_m \rho)$$

and so

$$\rho = \sum_m \text{tr}(M_m^\dagger M_m \rho) \left(\frac{M_m \rho M_m^\dagger}{\text{tr}(M_m^\dagger M_m \rho)} \right) = \sum_m M_m \rho M_m^\dagger$$

$\Rightarrow$ Mixture

$$\boxed{\rho = \sum_m M_m \rho M_m^\dagger}$$

* Density operator describes ensembles of Quantum states.

$\rightarrow$ Aim to move away from this interpretation

Thm: Characterization of density operators

An op ρ is the density op associated to some ensemble $\{\rho_i, |\psi_i\rangle\}$ iff it satisfies the conditions

(i) Trace $\text{Tr}(\rho) = 1$

(ii) Positivity ρ is a positive operator
 $\rho v \geq 0 \quad \forall v$

Proof: $\rho = \sum_i P_i |\psi_i\rangle \langle \psi_i|$

$\Rightarrow$ (i) $\text{Tr}(\rho) = \sum_i P_i \langle \psi_i | \psi_i \rangle = \sum_i P_i = 1$

(ii) $|\psi\rangle \in \mathcal{H}$,

$$\begin{aligned} \langle \psi | \rho | \psi \rangle &= \sum_i P_i \langle \psi | \psi_i \rangle \langle \psi_i | \psi \rangle \\ &= \sum_i P_i |\langle \psi | \psi_i \rangle|^2 \geq 0 \end{aligned}$$

$\Leftarrow$ Suppose $\exists \rho$ s.t. $\text{Tr}(\rho) = 1$ and $\langle \psi | \rho | \psi \rangle \geq 0$

Since ρ is positive it has a spectral decomp

$$\rho = \sum_i \lambda_i |i\rangle\langle i|, \quad \lambda_i \in \mathbb{R}^+$$

$$\text{Tr}(\rho) = 1 \Rightarrow \sum_i \lambda_i = 1$$

*Can define ρ to be the operator s.t. $\text{Tr}(\rho) = 1$

Postulate 1: Associate to any isolated physical system is $\mathcal{H}$ described by $\text{Tr}(\rho) = 1$. If a quantum state is in the state ρ_i with prob. p_i : $\sum_i p_i \rho_i$ system.

2: Evolution of a closed system

$$\rho' = U\rho U^\dagger \quad U \in U(\mathcal{H})$$

3: Measurements $\{M_m\}$ operators.

m : refers to measurement outcome.

If before the measurement the system is ρ

$$p(m) = \text{Tr}(M_m^\dagger M_m \rho)$$

$$\left[\begin{aligned} p(m|i) &= \langle \psi_i | M_m^\dagger M_m | \psi_i \rangle = \sum_j \langle \psi_i | M_m^\dagger | j \rangle \langle j | M_m | \psi_i \rangle \\ &= \sum_j \langle j | M_m | \psi_i \rangle \langle \psi_i | M_m^\dagger | j \rangle \\ &= \text{Tr}(M_m^\dagger M_m |\psi_i\rangle\langle\psi_i|) \\ p(m) &= \sum_i p(m,i) = \sum_i p(m|i) p(i) \\ &= \sum_i \text{Tr}(M_m^\dagger M_m |\psi_i\rangle\langle\psi_i|) p(i) \\ &= \text{Tr}(M_m^\dagger M_m \rho) \end{aligned} \right] \quad \left| \psi_i^m \right\rangle = \frac{M_m |\psi_i\rangle}{\sqrt{p(m|i)}}$$

Before: ρ

↓ measurement (M_m) :

$$\sum_m M_m^\dagger M_m = \mathbb{I}$$

After: $\rho_m = \frac{M_m \rho M_m^\dagger}{\text{Tr}(M_m^\dagger M_m \rho)} = \frac{M_m \rho M_m^\dagger}{p(m)}$

4: $\mathcal{H} = \mathcal{H}^{\otimes n}$: $\rho_1 \otimes \rho_2 \otimes \dots \otimes \rho_n$

* Best used when

i) Description of quantum systems whose states is not known

ii) Subsystems of composite quantum systems

Criteria for Pure/Mixed states: ρ be a density op

i) Then $\text{Tr}(\rho^2) \leq 1$

ii) $\text{Tr}(\rho^2) = 1$ if ρ is pure

Proof:

$$\begin{aligned} \text{i) } \text{Tr}(\rho^2) &= \sum_{i,j} p_i p_j \text{Tr}(|i\rangle\langle i| |j\rangle\langle j|) \\ &= \sum_{i,j} p_i^2 \text{Tr}(|i\rangle\langle i|) \\ &= \sum_i p_i^2 \end{aligned}$$

$$\text{Tr}(\rho^2) \geq 0$$

$$\bullet \sum p_i = 1 \Rightarrow p_i \leq 1 \quad \forall i$$

$$p_i^2 \leq p_i \leq 1$$

$$p_i^2 + p_j^2 \leq p_i + p_j \leq 1$$

$$\text{induction: } \sum_i p_i^2 \leq \sum_i p_i = 1$$

$$\Rightarrow \sum_i p_i^2 \leq 1$$

$$\Rightarrow 0 \leq \text{Tr}(\rho^2) \leq 1$$

$$\bullet \sum_i p_i^2 = 1 \Rightarrow \sum_i p_i^2 = \sum_i p_i = 1$$

$$p_i \leq 1 \Rightarrow p_i^2 \leq p_i \leq 1$$

Equality holds $\Rightarrow p_i = 1$ and only one such

case: $p_i = 1, i=0 \quad \forall j \neq i$

$\rho = |i\rangle\langle i|$: pure.

* Suppose $|a\rangle = \sqrt{\frac{3}{4}}|0\rangle + \sqrt{\frac{1}{4}}|1\rangle$

$$|b\rangle = \sqrt{\frac{3}{4}}|0\rangle - \sqrt{\frac{1}{4}}|1\rangle$$

$$\Rightarrow |4\rangle = \sqrt{\frac{1}{2}}|a\rangle + \sqrt{\frac{1}{2}}|b\rangle$$

$$\begin{aligned} \rho &= \sum_i \frac{1}{2} |4\rangle\langle 4| = \frac{1}{2} |a\rangle\langle a| + \frac{1}{2} |b\rangle\langle b| \\ &= \frac{3}{4} |0\rangle\langle 0| + \frac{1}{4} |1\rangle\langle 1| \end{aligned}$$

$\Rightarrow$ Two different ensembles give to the same density matrix

* The eigenvectors & eigenvalues of a density matrix

just indicate one of many possible ensembles

$\Rightarrow \rho$ gives us an equivalence class of ensembles.

Q: What class of states give rise to a particular ρ ?

Answer: $|\tilde{\psi}_i\rangle$: may not be normalized

$\Rightarrow |\psi_i\rangle$ generates the operator $\rho \equiv \sum_i |\psi_i\rangle\langle\psi_i|$

$$|\tilde{\psi}_i\rangle = \sqrt{p_i} |\psi_i\rangle$$

Thm: (Unitary freedom in the ensemble for density matrix)

$|\psi_i\rangle$ and $|\tilde{\psi}_i\rangle$ generate the same density matrix iff

$$|\psi_i\rangle = \sum_j u_{ij} |\tilde{\psi}_j\rangle$$

$$u_{ij} \in U(n, \mathbb{C})$$

$$\Rightarrow \rho = \sum_i p_i |\psi_i\rangle\langle\psi_i| = \sum_j q_j |\tilde{\psi}_j\rangle\langle\tilde{\psi}_j|$$

for normalized states $|\psi_i\rangle, |\tilde{\psi}_j\rangle$ iff

$$\sqrt{p_i} |\psi_i\rangle = \sum_j u_{ij} \sqrt{q_j} |\tilde{\psi}_j\rangle$$

* Reduced Density operator:

$\Rightarrow$ Subsystem description

Systems A and B:

• The entire system is described by ρ^{AB}

$$\rho^A = \text{Tr}_B(\rho^{AB}) \quad (\text{Reduced density op for A})$$

• Tr_B : partial trace

$$\text{Tr}_B(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|) \equiv |a_1\rangle\langle a_2| \text{Tr}(|b_1\rangle\langle b_2|)$$

$$\hookrightarrow \text{Tr}_B(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|) = \langle b_2 | b_1 \rangle |a_1\rangle\langle a_2|$$